



SIMATS
ENGINEERING



SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Crowd-Sourced Community Infrastructure Damage Mapping with Severity Prioritization

CSA0925-PROGRAMMING IN JAVA

Guided by :

Dr. Sivasankar C

Name of the student :

K.Dhilip (192365013)

Lithish Kumar R (192424169)

Mohammed Subahaan(192572165)

Domain

● Domain

- Smart City Infrastructure Management
- GIS-Based Civic Issue Monitoring
- AI-Assisted Infrastructure Analysis

● Project Summary

- Enables citizens to report infrastructure damage.
- Uses location, images and descriptions for analysis.
- Prioritizes issues based on severity and report density.
- Helps authorities manage repair activities.

● Target Users

- Citizens
- Municipal Authorities
- Maintenance Teams

Problem Statement

● Problem

- Infrastructure issues are often detected late.
- Manual inspections are time-consuming.
- Authorities lack real-time ground-level information.
- Critical issues may not receive priority.

● Impact

- Delayed repairs
- Increased maintenance cost
- Public inconvenience
- Reduced infrastructure safety

MOTIVATION & NEED

● Motivation

- Improve civic issue reporting.
- Reduce dependency on manual inspections.
- Identify high-priority damage quickly.

● Need

- Centralized reporting platform
- Location-based issue monitoring
- Severity-based prioritization
- Transparent repair tracking

● Social Impact

- Safer communities
- Faster response
- Better citizen participation

EXISTING SYSTEM / CURRENT PRACTICE

● Existing Approaches

- Manual infrastructure inspections
- Phone/email-based complaints
- Separate complaint management systems
- Basic GIS mapping systems
- AI-based road damage detection systems

● Limitations

- Delayed reporting
- Limited citizen participation
- No unified severity prioritization
- Limited repair-status visibility
- Manual resource allocation

RESEARCH / KNOWLEDGE GAP

● Identified Gap

- Existing systems often focus on detection or reporting alone.
- Limited integration of citizen-generated data and AI analysis.
- Severity and report density are not always combined for prioritization.
- Repair tracking is often separated from reporting.

Opportunity: Integrate reporting + severity analysis + priority heatmaps + authority action tracking in one platform.

PROPOSED SOLUTION

● Proposed System

- Citizens submit infrastructure damage reports.
- Reports include photos, descriptions and GPS locations.
- AI/ML analyzes damage indicators.
- 9-factor engine calculates priority.
- Heatmaps highlight critical areas.
- Authorities assign and track repair actions.
- Citizens receive status updates.

Key Benefit: From issue reporting to repair tracking in one integrated platform.

OBJECTIVES

Primary Objective: Develop a smart platform for community-based infrastructure damage reporting and severity prioritization.

● Specific Objectives

- Enable geotagged infrastructure reporting.
- Classify damage using AI/ML techniques.
- Calculate priority using multiple severity factors.
- Generate interactive priority heatmaps.
- Track authority repair actions.
- Provide status updates to citizens.

REQUIREMENTS & SPECIFICATIONS

● Functional Requirements

- User authentication
- Issue reporting
- Photo and GPS capture
- Severity classification
- Priority calculation
- Heatmap generation
- Repair task tracking
- Status updates

● Non-Functional Requirements

- Security
- Reliability
- Scalability
- Usability
- Fast response

● Hardware

- Laptop/Desktop
- Smartphone with GPS & Camera
- Internet connection

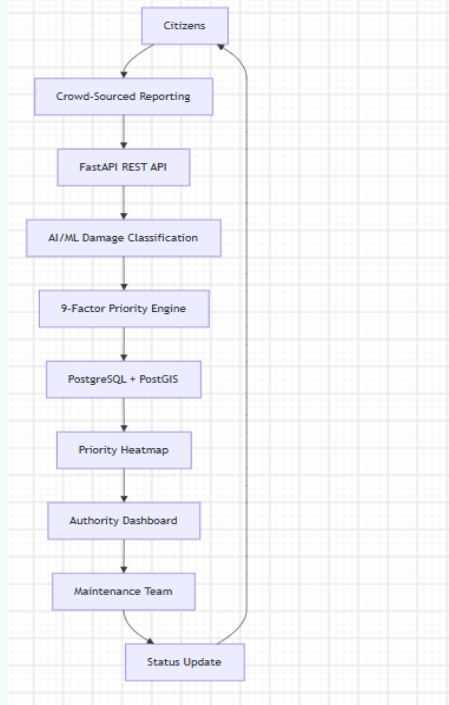
● Software

- Modern Web Browser
- Vercel (Deployment)
- Render
- Github

LITERATURE / TECHNOLOGY REVIEW

Author(s) & Year	Paper / Approach	Advantage	Disadvantage
F. Wan et al. (2022)	YOLO-LRDD: Lightweight road damage detection using improved YOLOv5s	Fast and lightweight detection	Focuses mainly on damage detection
Y. H. Zhang et al. (2023)	YOLO-LWNet: Lightweight YOLO-based road damage detection for mobile devices	Low computational cost and suitable for mobile devices	Limited to image-based detection
L. A. Silva et al. (2023)	Automated Road Damage Detection Using UAV Images and Deep Learning Techniques	Covers large areas using UAV images and deep learning	Requires UAV-based image collection
A. Behzadian et al. (2024)	Optimizing Road Damage Detection with YOLOv10	Improves detection efficiency and generalization	Mainly focuses on detection, not repair management
D. Arya et al. (2024)	ORDDC'2024: State-of-the-Art Solutions for Optimized Road Damage Detection	Compares advanced road-damage detection solutions	Concentrates on detection rather than end-to-end civic management

SYSTEM ARCHITECTURE



1

User Interface Layer

2

YOLO v12/Llama 4 scout

3

AI/ML & Priority Layer

4

Database & Geospatial Layer

5

Authority Action Layer

METHODOLOGY / WORKFLOW

1 Report

Citizen submits photo, location and description.

2 Validate

System validates user and report information.

3 Analyze

AI/ML identifies damage characteristics.

4 Prioritize

9-factor engine calculates priority.

5 Visualize

Heatmap displays high-priority locations.

6 Assign

Authority assigns repair work.

7 Track

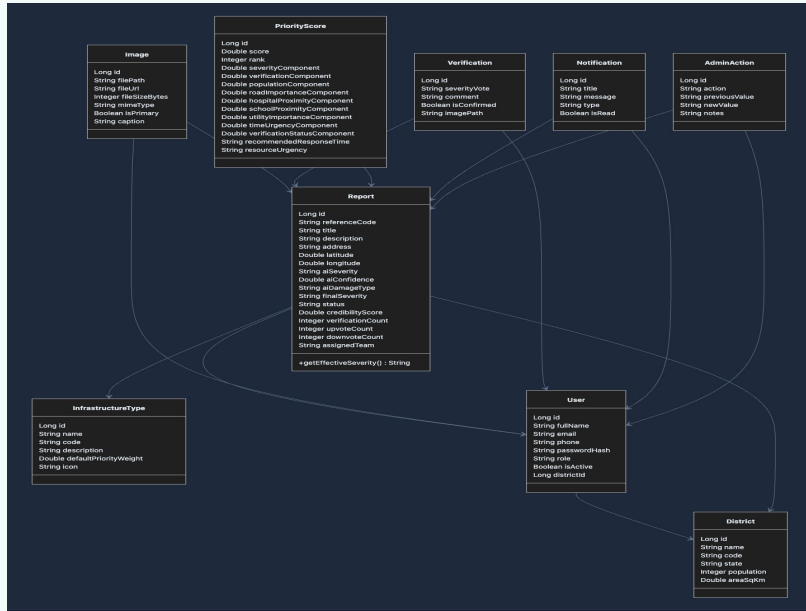
Maintenance team updates repair status.

8 Notify

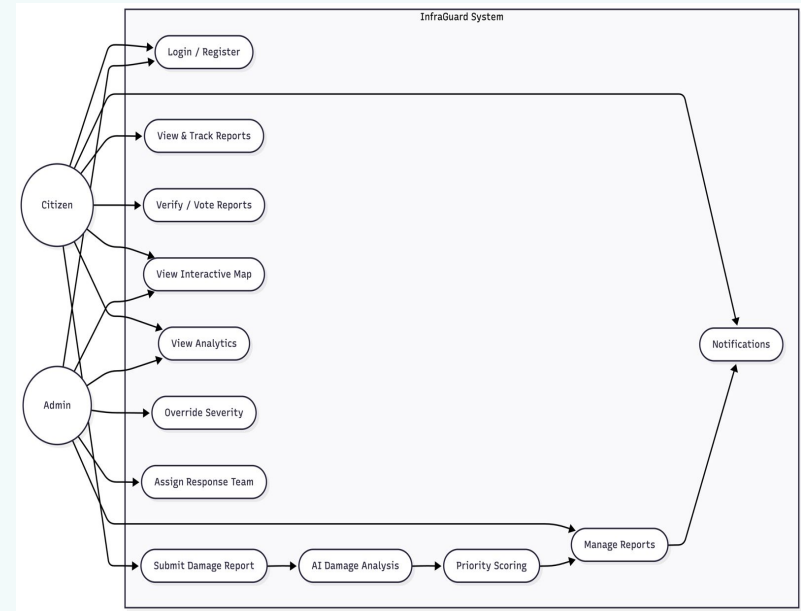
Citizen receives status information.

SYSTEM / PRODUCT DESIGN

CLASS DIAGRAM



USE CASE DIAGRAM



TECHNOLOGY STACK & TOOLS

● Frontend

- React 18
- Vite 5
- Tailwind CSS 3
- React Router 6
- Leaflet
- MarkerCluster
- Heatmap
- Chart.js
- Axios

● Backend

- Java 17
- Spring Boot
- Spring Data JPA
- Spring Security (JWT)
- Hibernate
- Maven
- BCrypt

● Database

- PostgreSQL 15
- PostGIS 3.3
- GeoAlchemy2
- Geospatial Indexes
- Database Triggers

● AI / ML

- OpenCV (Java)
- Weka
- D L 4 J (DeepLearning4J)
- ND4J
- Java ImageIO
- Hybrid Rule-based + ML Classifier
- 9-factor Priority Engine

● DevOps & Testing

- Docker
- Docker Compose
- nginx
- JUnit
- Vitest
- Testing Library

MODULE 1 DESIGN

Crowd-Sourced Reporting Module

● Purpose

- Collect infrastructure damage reports from citizens.

● Features

- User authentication
- Issue description
- Photo upload
- GPS location
- Report submission
- Report history

● Data Flow

- Citizen → Report Form → Photo/GPS → API → Database

● Output

- Stored and geotagged infrastructure report

MODULE 2 DESIGN

Severity Scoring & Heatmap Module

● Purpose

- Analyze damage severity and identify priority locations.

● Features

- Image preprocessing
- AI/ML damage classification
- Report-density analysis
- 9-factor priority calculation
- Heatmap generation

● Data Flow

- Report → AI/ML → Severity → Priority Engine → Heatmap

● Output

- Priority score
- Severity category
- Interactive heatmap

MODULE 3 DESIGN

Authority Action Tracking Module

● Purpose

- Manage and track infrastructure repair activities.

● Features

- View prioritized reports
- Assign repair tasks
- Track repair progress
- Update task status
- Provide citizen status updates

● Data Flow

- Authority → Assign Task → Maintenance Team → Status Update → Citizen

● Output

- Updated repair status
- Transparent action tracking

KEY ALGORITHM / TECHNICAL IMPLEMENTATION

● Hybrid Damage Classification

Process

- Image → Preprocessing → Feature Analysis → ML Classification → Damage Category

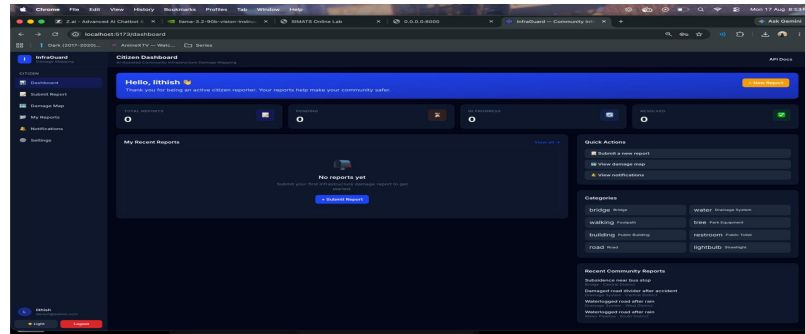
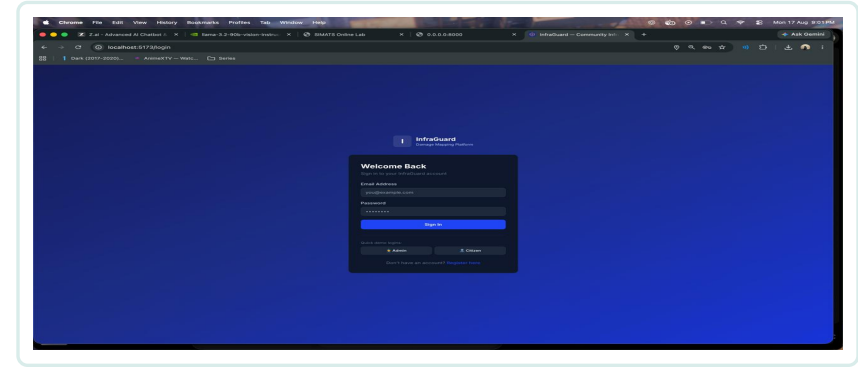
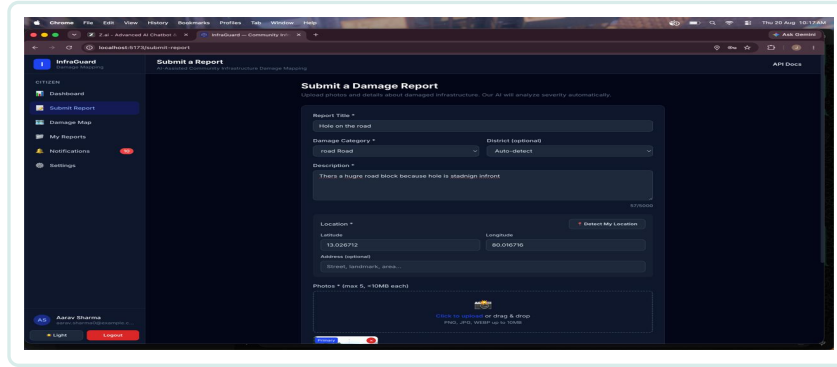
● Priority Engine

The system uses 9 factors to calculate issue priority:

- Damage Severity
- Report Density
- Location Importance
- Safety Impact
- Traffic Impact
- Public Usage
- Damage Extent

Final Output: Priority Score → Low / Medium / High / Critical

PROTOTYPE / WORKING MODEL



TESTING & VALIDATION

● Unit Testing

- Backend → JUnit
- Frontend → Vitest
- UI → Testing Library

● Integration Testing

- React ↔ Spring Boot REST API
- Spring Boot ↔ PostgreSQL/PostGIS
- Report → AI/ML → Priority Engine

● Test Cases

- Login validation
- Issue submission
- Photo upload
- Severity classification
- Priority calculation
- Status update

COMPARISON WITH EXISTING SOLUTIONS

Existing System	Proposed System
Manual reporting	Crowd-sourced reporting
Delayed detection	Faster issue reporting
Basic complaint handling	Severity-based prioritization
Limited location analysis	GIS-based heatmaps
Separate repair tracking	Integrated action tracking
Limited transparency	Citizen status updates

Key Improvement: Reporting + AI/ML + Priority + Heatmap + Repair Tracking in one platform.

CHALLENGES & ENGINEERING DECISIONS

Challenge	Solution
GPS/location accuracy	PostGIS geospatial processing
Image variations	OpenCV preprocessing
Damage classification	Hybrid ML + rule-based approach
Multiple map markers	Marker clustering
Priority calculation	9-factor priority engine
API security	JWT authentication
Large geospatial data	Spatial indexing

INDIVIDUAL CONTRIBUTIONS

● Team Member 1 – K.Dhilip

- Frontend development
- UI implementation
- Map integration

● Team Member 2 – Mohammed Subahaan

- Backend/API development
- Database integration
- Authentication

● Team Member 3 – Lithish Kumar R

- AI/ML implementation
- Severity classification
- Priority engine

● Team Member 3 – Lithish Kumar R

- Testing
- Documentation
- Deployment

PROJECT PROGRESS / MILESTONES

● Completed

- Requirement analysis
- System architecture
- Database design
- Three-module design
- Frontend development
- Backend API development
- AI/ML integration
- Priority engine
- Heatmap implementation
- Testing

Current Status — Project Implementation: 90% Completed



LIMITATIONS

- Depends on citizen participation.
- Incorrect reports may affect analysis.
- GPS accuracy may vary.
- Image quality affects classification.
- ML performance depends on training data.
- Internet connection is required.
- Current system focuses on selected infrastructure damage types.

FUTURE ENHANCEMENTS

- Mobile application
- Real-time notifications
- Advanced deep-learning models
- Automatic duplicate-report detection
- More infrastructure damage categories
- Predictive maintenance
- Integration with municipal systems
- Large-scale city deployment
- Multilingual support

CONCLUSION & PROJECT OUTCOMES

● Conclusion

- Developed a community-driven infrastructure monitoring platform.
- Enables geotagged issue reporting.
- Uses AI/ML for damage classification.
- Uses a 9-factor engine for priority calculation.
- Generates interactive priority heatmaps.
- Tracks authority repair actions.

● Project Outcomes

- Faster issue identification
- Better repair prioritization
- Improved resource allocation
- Greater transparency
- Increased citizen participation

REFERENCES

1. “Deep Learning Approaches for Road Damage Detection Using YOLOv8 and Custom Convolutional Neural Network,” IEEE Conference Publication, 2025. IEEE Xplore
2. “Evaluation on Deep Learning Algorithms for Real-Time Road Damage Detection,” IEEE Conference Publication, 2025. IEEE Xplore
3. “Automated Road Damage Detection Using UAV Images and Deep Learning Techniques,” IEEE Journals & Magazine. IEEE Xplore
4. “Implementing Crowdsourcing in Smart Government: An IT Perspective Review,” IEEE Conference Publication, 2024. IEEE Xplore
5. “Smartphone-based Road Condition Sensing using Machine Learning techniques,” IEEE Conference Publication. IEEE Xplore
6. “AI assisted pothole detection and depth estimation,” IEEE Conference Publication.

GitHub link:

<https://github.com/lithish-source/infraguard-working-version>

Hosting Link

<https://infraguard-working-version.onrender.com>

**THANK
YOU**

shutterstock.com · 2601615063

GPS Photo

